LongBench v2: Towards Deeper Understanding and Reasoning on Realistic Long-context Multitasks

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https://longbench2.github.io

Abstract

This paper introduces LongBench v2, a benchmark designed to assess the ability of LLMs to handle long-context problems requiring *deep understanding and reasoning* across real-world multitasks. LongBench v2 consists of 503 challenging multiple-choice questions, with contexts ranging from 8k to 2M words, across six major task categories: single-document QA, multi-document QA, long in-context learning, long-dialogue history understanding, code repository understanding, and long structured data understanding. To ensure the breadth and the practicality, we collect data from nearly 100 highly educated individuals with diverse professional backgrounds. We employ both automated and manual review processes to maintain high quality and difficulty, resulting in human experts achieving only 53.7% accuracy under a 15-minute time constraint. Our evaluation reveals that the best-performing model, when directly answers the questions, achieves only 50.1% accuracy. In contrast, the o1-preview model, which includes longer reasoning, achieves 57.7%, surpassing the human baseline by 4%. These results highlight the importance of enhanced reasoning ability and scaling inference-time compute to tackle the long-context challenges in LongBench v2.

1 Introduction

Over the past year, research and products on long-context large language models (LLMs) have made remarkable progress: in terms of context window length, advancing from the initial 8k to the current 128k and even 1M tokens (OpenAI, 2024c; Anthropic, 2024; Reid et al., 2024; GLM et al., 2024); and achieving promising performance on long-context benchmarks. However, beneath these advancements lies an urgent and practical question:

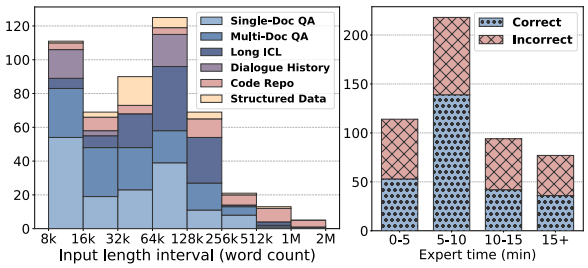


Figure 1: Length distribution (left) and human expert solving time distribution (right) of LongBench v2.

Do these models truly comprehend the long texts they process, i.e., are they capable of deeply understanding, learning, and reasoning based on the information contained in these long texts?

Critically, existing long-context understanding benchmarks (Bai et al., 2024b; Zhang et al., 2024d; Hsieh et al., 2024) fail to reflect the long-context LLMs' *deep* understanding capabilities across diverse tasks. They often focus on extractive questions, where answers are directly found in the material, a challenge easily handled by modern long-context models and RAG systems, as evidenced by their perfect recall in the Needle-in-a-Haystack test (Kamradt, 2023). Furthermore, many of these benchmarks rely on synthetic tasks, which limits their applicability to real-world scenarios, and their adopted metrics like F1 and ROUGE are unreliable.

To address these issues, we aim to build a benchmark with the following features: (1) **Length**: Context length ranging from 8k to 2M words, with the majority under 128k. (2) **Difficulty**: Challenging enough that even human experts, using search tools within the document, cannot answer correctly in a short time. (3) **Coverage**: Cover various realistic scenarios. (4) **Reliability**: All in a multiple-choice question format for reliable evaluation.

With the above goal in mind, we present *LongBench v2*. LongBench v2 contains 503 multiple-choice questions and is made up of 6 major task cat-

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egories and 20 subtasks to cover as many realistic deep comprehension scenarios as possible, including *single-document QA*, *multi-document QA*, *long in-context learning*, *long-dialogue history understanding*, *code repository understanding*, and *long structured data understanding* (detailed in Table 1). All the test data in LongBench v2 are in English, and the length distribution of each task category is shown on the left of Figure 1.

To ensure the quality and difficulty of test data, we combine automated and manual reviews during data collection. We first recruit 97 data annotators with diverse academic backgrounds and grades from top universities and then select 24 data reviewers from this group. Annotators provide data including long documents, questions, options, answers, and evidence. We then leverage three long-context LLMs for an automated review, where a question is considered too easy if all three LLMs answer it correctly. Data passing the automated review are assigned to the reviewers, who answer the questions and determine whether the questions are appropriate (meet our requirements) and if the answers are correct. In our criteria, a qualified data point should have (1) an appropriate question with an objective, correct answer; (2) sufficient difficulty, such that all three LLMs cannot answer correctly at the same time, and the human reviewer cannot answer correctly within 3 minutes, even with searching tools within the document. If data do not meet these criteria, we request modifications from the annotator. We also set length and difficulty incentives to encourage longer and harder test data. Figure 1 (right) visualizes the distribution of expert solving times along with human accuracy.

Overall, our data shows a median word count of 54k and an average of 104k words. Human experts are able to achieve an accuracy of only 53.7% within 15 minutes, compared to 25% accuracy with random guessing, highlighting the challenging nature of the test. In the evaluation, the best-performing model achieves only 50.1% accuracy when directly outputting the answer. In contrast, the o1-preview model, which incorporates longer reasoning during inference, reaches 57.7%, surpassing human experts. This implies that LongBench v2 places greater demands on the reasoning ability of current models, and incorporating more inference-time thinking and reasoning appears to be a natural and crucial step in addressing such long-context reasoning challenges. We hope LongBench v2 will accelerate the exploration of how scaling inference-time compute will affect deep understanding and reasoning in long-context scenarios.

2 Related Work

We divide existing long-context benchmarks for LLMs into two types. The first consists of comprehensive benchmarks that combine multitasks such as QA, retrieval, and summarization. Sorted by publication date, these benchmarks include ZeroSCROLLS (Shaham et al., 2023), L-Eval (An et al., 2024), LongBench (Bai et al., 2024b), BAMBOO (Dong et al., 2024), LooGLE (Li et al., 2024b), ∞-bench (Zhang et al., 2024d), Ruler (Hsieh et al., 2024), and HELMET (Yen et al., 2025). It is noteworthy that most of these multitask benchmarks were proposed last year, which corresponds to the thrive of long-context LLMs, whose context length has been extended to 128k tokens or more (Anthropic, 2024; OpenAI, 2024c; Reid et al., 2024; GLM et al., 2024; Dubey et al., 2024) through continual training (Xiong et al., 2024; Fu et al., 2024; Bai et al., 2024a; Gao et al., 2024).

The other category of long-context benchmarks is more targeted, evaluating models on specific types of long-context tasks, including document QA (Kočiskỳ et al., 2018; Dua et al., 2019; Dasigi et al., 2021; Pang et al., 2022; Wang et al., 2024a), summarization (Zhong et al., 2021; Huang et al., 2021; Wang et al., 2022), retrieval and attributing (Kamradt, 2023; Kuratov et al., 2024; Song et al., 2025; Laban et al., 2024; Zhang et al., 2024b; Vodrahalli et al., 2024; Krishna et al., 2025), conversation (Bai et al., 2024a), coding (Liu et al., 2024a; Bogomolov et al., 2024), many-shot learning (Agarwal et al., 2024), and long-text generation (Bai et al., 2025; Wu et al., 2025b; Liu et al., 2024b; Que et al., 2024).

In our view, existing long-context benchmarks generally have the following issues: (1) *Lack of deep reasoning*: While a few benchmarks contain longer examples of around 100k, most of these data have not been human-examined, and many of these samples can be solved through shallow understanding such as retrieval, thus failing to reflect a model's deep reasoning capabilities. (2) *Unreliable metrics*: Many datasets use metrics like ROUGE and F1 for evaluation, which are known to be unreliable (Novikova et al., 2017). Additionally, some datasets adopt LLM-as-a-judge (Zheng et al., 2023; Li et al., 2024a) for evaluation, which can be costly

Dataset	Source	#data	Length	Expert Acc	Expert Time*
I. Single-Document QA		175	51k	55%	8.9 min
Academic	Paper, textbook	44	14k	50%	7.3 min
Literary	Novel	30	72k	47%	8.5 min
Legal	Legal doc	19	15k	53%	13.1 min
Financial	Financial report	22	49k	59%	9.0 min
Governmental	Government report	18	20k	50%	9.5 min
Detective	Detective novel	22	70k	64%	9.3 min
Event ordering	Novel	20	96k	75%	9.4 min
II. Multi-Document QA		125	34k	36%	6.1 min
Academic	Papers, textbooks	50	27k	22%	6.1 min
Legal	Legal docs	14	28k	64%	8.8 min
Financial	Financial reports	15	129k	40%	7.0 min
Governmental	Government reports	23	89k	22%	6.0 min
Multi-news	News	23	15k	61%	5.3 min
III. Long In-context Learning		81	71k	63%	8.3 min
User guide QA	Electronic device, software, instrument	40	61k	63%	9.9 min
New language translation	Vocabulary book (*Kalamang*, *Zhuang*)	20	132k	75%	5.4 min
Many-shot learning	Multi-class classification task	21	71k	52%	8.0 min
IV. Long-dialogue History Understanding		39	25k	79%	8.2 min
Agent history QA	LLM agents conversation	20	13k	70%	8.3 min
Dialogue history QA	User-LLM conversation	19	77k	89%	6.5 min
V. Code Repository Understanding		50	167k	44%	6.4 min
Code repo QA	Code repository	50	167k	44%	6.4 min
VI. Long Structured Data Understanding		33	49k	73%	6.4 min
Table QA	Table	18	42k	61%	7.4 min
Knowledge graph reasoning	KG subgraph	15	52k	87%	6.2 min

Table 1: Tasks and data statistics in LongBench v2. 'Source' denotes the origin of the context. 'Length' is the *median* of the number of words. 'Expert Acc' and 'Expert Time' refer to the average accuracy and the *median* time spent on answering the question by human experts. *: We allow human experts to respond with "I don't know the answer" if it takes them more than 15 minutes. As a result, most expert times are under 15 minutes, but this doesn't necessarily mean that the questions are fully answered within such a time.

and may introduce biases in their assessments (Bai et al., 2024c; Wang et al., 2024b; Ye et al., 2025). To construct a more challenging, reliable, and comprehensive long-context benchmark, we employ a uniform multiple-choice format and manually verify each data point to ensure it meets the required level of difficulty.

3 LongBench v2: Task and Construction

Our design principle focuses on four aspects: (1) The context should be sufficiently long to cover scenarios ranging from 8k to 2M words, with a relatively even distribution across texts up to 128k words. (2) The question should be challenging, requiring the model to deeply understand the context to answer. It should avoid questions that can be answered based on memory or those where the answer can be directly extracted from the context. (3) The data should cover a wide range of real-world long-context scenarios and reflect the model's holistic ability to reason, apply, and analyze information drawn from the lengthy text. (4) The data should be in English and in a multiple-choice question format, containing a long text, a question, four choices, a groundtruth answer, and an evidence. Distractors should be included to prevent the model from guessing the correct answer based on option patterns.

3.1 Task Overview

Based on the testing scenarios and the types and sources of long texts, we propose six major task categories and further divide them into 20 subtasks. We introduce the tasks included in LongBench v2 in the following. A list of task statistics and detailed descriptions can be found in Table 1 and Appendix B.

Single-Doc QA. We integrate subtask categories from previous datasets (Bai et al., 2024b; An et al., 2024) and expand them to include QA for *academic*, *literary*, *legal*, *financial*, and *governmental*

documents. Considering that *detective* QA (Xu et al., 2024) requires in-depth reasoning based on case background, we introduce such a task that requires identifying the killer or motive based on information provided in detective novels. We also include *Event ordering*, where the goal is to order minor events according to the timeline of a novel.

Multi-Doc QA. To distinguish from single-doc QA, multi-doc QA requires answers drawn from multiple provided documents. Besides the categories in single-doc QA, multi-doc QA also includes *multi-news QA*, which involves reasoning across multiple news articles, events, and timelines.

Long In-context Learning. Learning from a long context, such as acquiring new skills, requires the ability to comprehend and reason based on that context. Hence, we consider it as a major category of tasks. LongBench v2 includes several key tasks, including *User guide QA*, which answers questions with information learnt from user guides for electronic devices, software, etc.; *New language translation* (Tanzer et al., 2024; Zhang et al., 2024a), which involves learning to translate an unseen language from a vocabulary book; *Many-shot learning* (Agarwal et al., 2024), which involves learning to label new data from a handful of examples.

Long-dialogue History Understanding. LLMs, as more intelligent chatbots or agents, require enhanced memory capabilities to handle longer histories. Therefore, we integrate long-dialogue history understanding tasks to test whether LLMs can handle information from long conversation histories. These tasks are divided into two subtasks based on the source of the conversation history: one involving the history of interactions between multiple LLM agents, i.e., *Agent history QA* (tse Huang et al., 2025), and the other involving the dialogue history between a user and an LLM acting as an assistant, i.e., *Dialogue history QA* (Wu et al., 2025a).

Code Repository Understanding. Code repository contains long code content, and question answering over a code repository requires understanding and reasoning across multiple files, making it a common yet challenging long-context task.

Long Structured Data Understanding. In addition to textual data, much information is presented in structured forms, so we introduce the long structured data QA task to test the LLM's understanding of long structured data, including reasoning on long tables, i.e., *Table QA* (Zhang et al., 2024c), and answering complex queries on knowledge graphs (KGs), i.e., *Knowledge graph reasoning* (Cao et al., 2022; Bai et al., 2023). We anonymize the entities in the KG to prevent the model from directly deriving the answers through memorization.

3.2 Data Collection

To collect high-quality and challenging data for long-context tasks, we hire 97 annotators who are either holding or pursuing a bachelor's degree from top universities and are proficient in English, with detailed statistics shown in Appendix C.2. We also select 24 professional human experts based on their major and year of study for conducting manual reviews. Figure 2 illustrates the overall pipeline of our data collection process, which consists of five steps: document collection, data annotation, automated review, manual review, and data revision (optional). We develop an online annotation platform to implement this pipeline, with further details provided in Appendix C.1.

Step 1: Document Collection. Unlike previous benchmarks (Bai et al., 2024b; An et al., 2024), where long documents are pre-defined or synthesized by the benchmark designers, we aim to gather documents that reflect more diverse scenarios and are more likely to be used in everyday contexts. To achieve this, we ask annotators to upload one or multiple files they have personally read or used, such as research papers, textbooks, novels, etc., according to the task type. Our platform first converts the uploaded files into plain text using tools such as `PyMuPDF`. The input documents then undergo two automatic checks. If the length is less than 8,192 words, it is rejected as too short. Documents with a high overlap with previous annotations are also rejected to ensure diversity.

Step 2: Data Annotation. During data annotation, the annotator is tasked with proposing a multiple-choice question based on their submitted documents. The question should be accompanied with four choices, a groundtruth answer, and the supporting evidence. We provide the annotators with a detailed question design principle that specifies our requirement (Appendix C.3). To summarize, the following types of questions should be avoided: (1) *Counting questions*: Avoid questions that require counting large numbers. (2) *Simple retrieval questions*: Do not ask basic information retrieval questions, as these are too easy for modern LLMs (Song et al., 2025). (3) *Overly professional questions*: Questions should not demand extensive external knowledge; they should rely on minimal

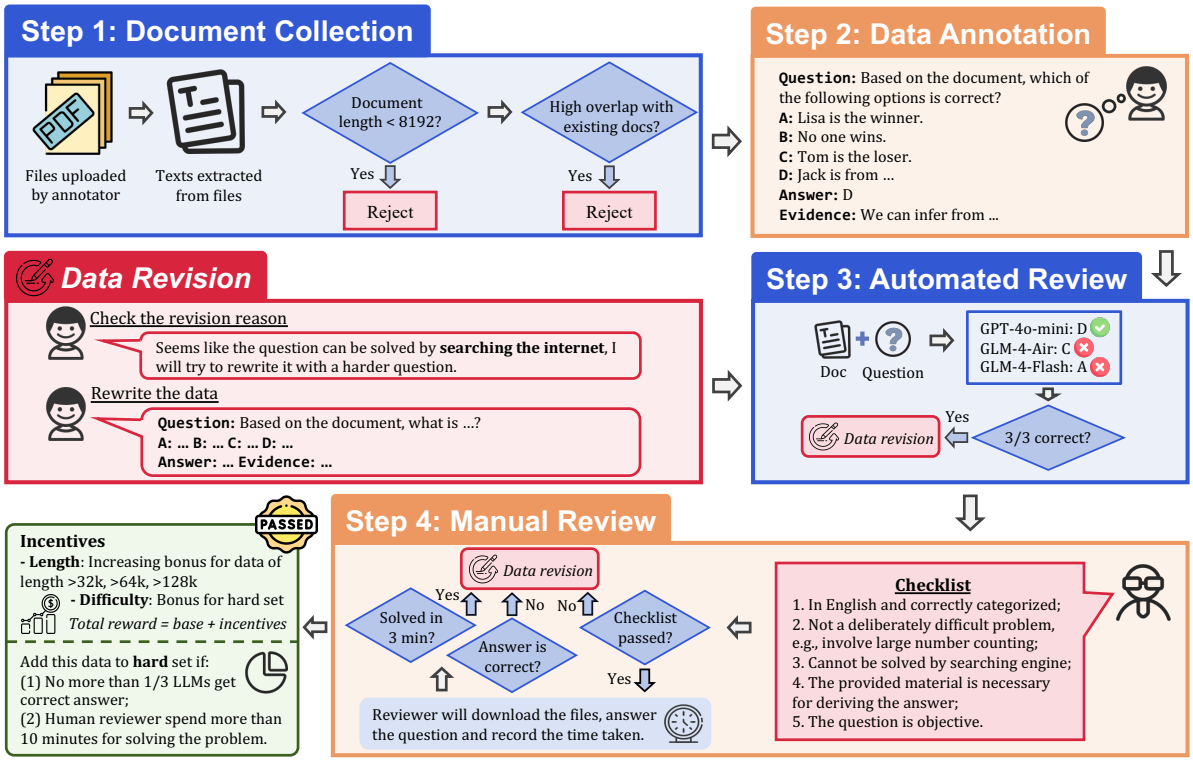


Figure 2: Data collection pipeline of LongBench v2. The annotator first uploads the document(s) and proposes a multiple-choice question based on the content. After that, automated and manual reviews will be conducted to ensure the data meets our requirements. Only data that passes these reviews is eligible for annotation rewards, meaning the annotator must revise the data until it passes all review stages. More details are in section 3.2.

expertise. (4) *Tricky questions*: Do not create questions that are deliberately difficult; the goal is to keep the questions natural and straightforward.

Step 3: Automated Review. Upon submission, each question undergoes an initial automated review process to ensure it is not too easy. We employ three fast and powerful LLMs with a 128k context length to answer the questions: GPT-4o-mini (OpenAI, 2024a), GLM-4-Air, and GLM-4-Flash. Inputs that exceed the context length are truncated from the middle. If all three LLMs answer the question correctly, it is considered too easy. In such cases, annotators will be required to revise the question and choices to increase its difficulty.

Step 4: Manual Review. Data passing the automated review is sent to a human expert for manual review. Our manual review serves two purposes: first, to filter out unqualified questions and data with incorrect answers; second, to establish a human baseline while also determining the difficulty of the questions and filter out those that are too easy (i.e., questions that humans can answer correctly in a short amount of time). In practice, the reviewer first goes through a checklist to determine whether the question meets the specified requirements (outlined in Appendix C.3). Next, the reviewer downloads the raw document files and attempts to answer the question. The reviewer is encouraged to use searching tools within the files to solve the problem more promptly. Once a choice is submitted, the reviewer can view the groundtruth answer and the evidence provided by the annotators. The reviewer will then decide whether the answer is objective and fully correct. Our platform tracks the time spent on each question, and if the human expert answers correctly within 3 minutes, the question will be considered too easy, demanding a revision from its annotator. Since answering some questions may require spending several hours reading the material, which implies a significant review time cost, we allow human experts to respond with "I don't know the answer" after 15 minutes.

Data Revision. As mentioned above, questions deemed unqualified during either automated or manual review will require revision by its annotator. We set up a separate page in our platform for annotator to track their rejected data. For each rejected data, we provide the annotator with a reason for the rejection, classified into three categories: (1) *Illegal question*: Rejected by human reviewers due

to the question being unqualified, (2) *Insufficient difficulty*: Rejected by automated review or due to human reviewer answering the question correctly within 3 minutes, and (3) *Wrong answer*: Rejected by human reviewers. Based on this feedback, annotators will refine their data until it passes the review process. To avoid wasting too much manual resources on low-quality data, we will terminate the review-revision cycle if the data has been revised more than five times without passing.

Mechanism Design. To incentivize annotators to provide high-quality, challenging, and longer test data, our reward mechanism is set as follows. First, annotators can receive a base reward of `100 CNY` only if the data passes the review process; no reward is given for data that does not pass. To encourage annotators to provide longer data, we offer additional length rewards of `20`, `40`, and `50 CNY` for passed data in the length ranges $(32k, 64k]$, $(64k, 128k]$, and over $128k$, respectively (in word count). To motivate annotators to provide more difficult data, we define *hard* set data as data where at least two out of three models do not answer correctly in automated review and the human reviewer is unable to solve it within 10 minutes; all other data is considered *easy* data. For hard data, annotators can earn an additional difficulty reward of `50 CNY`. Each human expert is rewarded `25 CNY` for reviewing each piece of data. We also conduct random checks on their reviews, and any human expert whose reviews repeatedly fail these checks will have all of their reviewing rewards revoked.

3.3 Data Verification

For a final check, we sample 70 test data and invite our authors to verify their correctness and whether they are Google-proofed (Rein et al., 2024).

Correctness. Check the selected answer based on the provided evidence to determine if it is correct, with all other options being incorrect. An answer is also deemed incorrect if there is any controversy, ambiguity, or reliance on subjective judgment.

Google-proof. Search for the answer to the question on the internet (Google). The data is considered Google-proof if the answer cannot be found within 15 minutes of searching.

Through our verification, we find that `68/70` of the data are completely correct, and `67/70` are Google-proofed. Therefore, we estimate that the error rate of our data is around 3%, and the majority of the questions cannot be answered by memorizing existing data on the internet. We review all the data to ensure that it does not contain any sensitive information related to privacy or copyrights.

3.4 Data Statistics

We categorize the 503 data entries in Longbench v2 based on their difficulty, length, and task types. According to the difficulty criteria defined in the previous section, 192 are classified as "Easy", while 311 are deemed "Hard". Based on word count, the data is divided into three groups: "Short" (<32k), "Medium" (32k-128k), and "Long" (>128k), containing 180, 215, and 108 entries, respectively, exhibiting a relatively balanced distribution. For the data distribution across task types, please see Table 1. Also, the questions with answers A, B, C, and D account for approximately 19%, 25%, 30%, and 26% of the total, respectively, showing that the distribution of answers across the four options is relatively even. We also analyze the proportion of data submissions rejected during manual review and find that 4% of the submissions are rejected for *illegal question*; 7% are rejected for *insufficient difficulty*; and 4% are rejected for *wrong answer*.

4 Evaluation

4.1 Baselines

Setup. We evaluate 10 open-source LLMs, all of which have a context window size of 128,000 tokens, along with 9 frontier proprietary LLMs. We apply middle truncation as described in Bai et al. (2024b) for sequences exceeding the model's context window length. Given the complex reasoning required by our test data, we adopt two evaluation settings: zero-shot and zero-shot + CoT. Following Rein et al. (2024), in the CoT setting, the model is first prompted to generate a chain of thought (Wei et al., 2022), after which it is asked to produce the final answer based on the chain of thought. For details on reproducing our results, please refer to Appendix D. For a fair comparison, the Qwen2.5 series models are evaluated without YaRN (Peng et al., 2024). Their performance when combining YaRN are provided in Table 4. The code is available at `https://github.com/THUDM/LongBench`.

Results. We report the evaluation results along with human expert performance in Table 2. The results under the CoT evaluation setting are highlighted with a gray background, while the highest scores among open-source models and proprietary models are in bold. The results indicate that LongBench v2 presents a significant challenge to the

Model	Overall		Difficulty				Length (<32k; 32k-128k; >128k)◇					
	Overall		Easy		Hard		Short		Medium		Long	
Open-source models												
GLM-4-9B-Chat	30.2	30.8	30.7	34.4	29.9	28.6	33.9	35.0	29.8	30.2	25.0	25.0
Llama-3.1-8B-Instruct	30.0	30.4	30.7	36.5	29.6	26.7	35.0	34.4	27.9	31.6	25.9	21.3
Llama-3.1-70B-Instruct	31.6	36.2	32.3	35.9	31.2	36.3	41.1	45.0	27.4	34.0	24.1	25.9
Llama-3.3-70B-Instruct	29.8	36.2	34.4	38.0	27.0	35.0	36.7	45.0	27.0	33.0	24.1	27.8
Llama-3.1-Nemotron-70B-Inst.	31.0	35.2	32.8	37.0	29.9	34.1	38.3	46.7	27.9	29.8	25.0	26.9
Qwen2.5-7B-Instruct	27.0	29.8	29.2	30.7	25.7	29.3	36.1	35.6	23.7	26.5	18.5	26.9
Qwen2.5-72B-Instruct	**39.4**	38.8	**43.8**	42.2	**36.7**	**36.7**	**44.4**	**50.0**	**34.0**	28.8	**41.7**	**39.8**
Mistral-Large-Instruct-2407	26.6	33.6	29.7	34.4	24.8	33.1	37.8	41.1	19.5	31.2	22.2	25.9
Mistral-Large-Instruct-2411	34.4	**39.6**	38.0	**43.8**	32.2	37.0	41.7	46.1	30.7	**34.9**	29.6	38.0
c4ai-command-r-plus-08-2024	27.8	31.6	30.2	34.4	26.4	29.9	36.7	39.4	23.7	24.2	21.3	33.3
Proprietary models												
GLM-4-Plus	44.3	46.1	47.4	52.1	42.4	42.4	50.0	53.3	46.5	44.7	30.6	37.0
GPT-4o-mini-2024-07-18	29.3	32.4	31.1	32.6	28.2	32.2	31.8	34.8	28.6	31.6	26.2	29.9
GPT-4o-2024-08-06	50.1	51.2	57.4	57.9	45.6	47.1	53.3	53.9	52.4	**50.7**	40.2	47.7
GPT-4o-2024-11-20	46.0	51.4	50.8	54.2	43.0	49.7	47.5	59.6	47.9	48.6	39.8	43.5
o1-mini-2024-09-12	37.8	38.9	38.9	42.6	37.1	36.6	48.6	48.9	33.3	32.9	28.6	34.3
o1-preview-2024-09-12	**57.7**	**56.2**	**66.8**	58.9	**52.1**	**54.6**	**62.6**	**64.6**	**53.5**	50.2	**58.1**	**54.3**
Claude-3.5-Sonnet-20241022	41.0	46.7	46.9	55.2	37.3	41.5	46.1	53.9	38.6	41.9	37.0	44.4
Gemini-Exp-1206	49.3	52.5	52.9	**61.5**	47.1	47.1	53.9	55.6	47.1	49.5	45.8	53.3
Gemini-2.0-Flash-Exp	47.4	51.1	51.0	58.3	45.2	46.6	48.9	57.2	47.7	45.3	44.4	52.3
*Human**	53.7		100		25.1		47.2		59.1		53.7	

Table 2: Evaluation results (%) on LongBench v2. Results under CoT prompting are highlighted with a gray background. Note that random guessing yields a baseline score of 25%. To account for model responses and human responses that do not yield a valid choice, we report the *compensated* results in Table 5, where these cases are counted towards the accuracy with a random probability of 25%. *: The human expert's accuracy is based on their performance within a 15-minute time limit, after which they are allowed to respond with "I don't know the answer". This occurred for 8% of the total test data. ◇: Models do not show lower scores on subsets with longer length ranges because the distribution of tasks differs significantly across each length range (Figure 1).

current model—The best-performing o1-preview model achieves only 57.7% accuracy, which is 4% higher than the performance of human experts under a 15-minute time limit. Additionally, the scaling law effect on our benchmark is striking: smaller models such as GLM-4-9B-Chat, Qwen2.5-7B-Instruct, and GPT-4o-mini perform poorly in our tests that require deep understanding and reasoning over long contexts, with accuracy around 30%. In contrast, their larger counterparts like GLM-4-Plus, Qwen2.5-72B-Instruct, and GPT-4o show a notable improvement, achieving overall accuracy around or above 40%. Similar to reasoning tasks in mathematics and coding (Wei et al., 2022; Sprague et al., 2025; OpenAI, 2024b), we also find that incorporating explicit reasoning in the model's responses significantly improves its performance in our long-context reasoning tests. This includes the use of CoT, which results in an average 3.4% improvement for open-source models. Additionally, scaling test-time compute with longer reasoning thought shows further improvements, with o1-preview vs. GPT-4o (+7.6%) and o1-mini vs. GPT-4o-mini (+8.5%). From the performance across different length intervals, compared to human, the models perform best on data <32k (Short), with the best-performing model surpassing human performance by 15.4%. However, even the top model shows a 5.6% performance gap compared to human accuracy in the 32k-128k data length range. This highlights the importance of developing methods to maintain strong reasoning capabilities under longer contexts.

To better distinguish the capability of the models across tasks, we present the performance charts of several representative models across tasks in Figure 3. We find that the performance gap between LLMs and humans is largest on long structured data understanding tasks, whereas, on single-doc and multi-doc QA tasks, the models perform at par with or even surpass human levels. We hypothesize that this is because the models have seen much more document-type data compared to long structured data during long context training, resulting

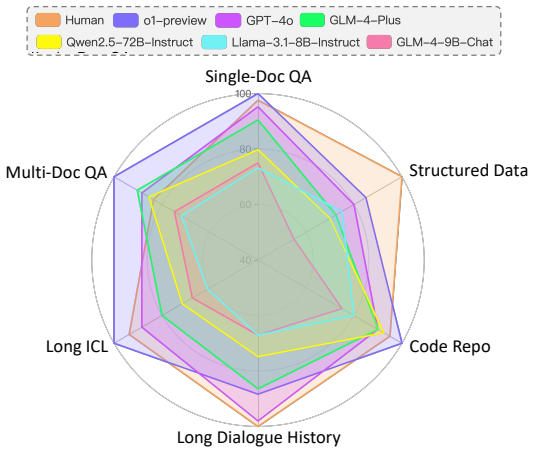


Figure 3: Average scores across tasks, normalized by the highest score on each task. All scores are evaluated in the zero-shot + CoT setting, except for o1-preview, since it latently performs CoT under zero-shot prompting.

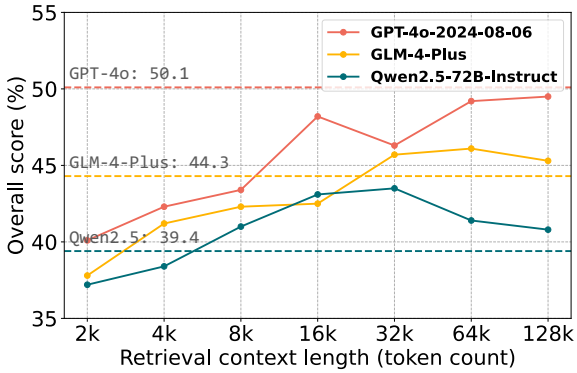


Figure 4: RAG performance across different context lengths, varied by including the top 4, 8, 16, 32, 64, 128, and 256 chunks of 512 tokens. The horizontal line show the overall score of each model without RAG at a full context length of 128k tokens.

in poorer understanding of the latter. Compared to GPT-4o, we observe that through integrating more thinking steps during inference, o1-preview shows superior performance on multi-doc QA, long in-context learning, and code repository understanding tasks, with a substantial lead over other models.

4.2 Retrieval-Augmented Baselines

Based on recent studies (Jiang et al., 2024; Jin et al., 2025; Leng et al., 2024), we explore incorporating retrieval-augmented generation (RAG, Lewis et al. (2020)) into long-context LLM and evaluate its performance on LongBench v2. We first split the long context into chunks of 512 tokens with GLM-4-9B tokenizer. Then, we use Zhipu Embedding-3 to encode the query, i.e., the concatenation of the question and choices, and the chunks, and sort

Model	Avg	I	II	III	IV	V	VI
GLM-4-9B-Chat	30.2	30.9	27.2	33.3	38.5	28.0	24.2
w/o context	26.2	30.9	21.6	18.5	30.8	34.0	21.2
Llama-3.1-8B-Inst.	30.0	34.9	30.4	23.5	17.9	32.0	30.3
w/o context	25.8	31.4	26.4	24.7	23.1	22.0	6.1
Qwen2.5-72B-Inst.	39.4	40.6	35.2	42.0	25.6	50.0	42.4
w/o context	30.0	33.7	31.2	25.9	28.2	34.0	12.1
GLM-4-Plus	44.3	41.7	42.4	46.9	51.3	46.0	48.5
w/o context	27.6	33.7	27.2	25.9	10.3	38.0	6.1
GPT-4o	50.1	48.6	44.0	58.0	46.2	56.0	51.5
w/o context	33.1	40.0	25.6	32.1	38.5	34.0	18.2

Table 3: Scores (%) across 6 tasks: *I. Single-Doc QA*, *II. Multi-Doc QA*, *III. Long ICL*, *IV. Dialogue History*, *V. Code Repo*, and *VI. Structured Data*.

the chunks based on embedding similarity. During evaluation, we retrieve the top-N most similar chunks and concatenate them in their original order to form the context input for the model. The model is then prompted to answer the question in a zero-shot setting. For each evaluated model, we take $N = 4, 8, 16, 32, 64, 128$, and 256, and the evaluation results form a curve presented in Figure 4.

We observe that Qwen2.5 and GLM-4-Plus show no significant improvement as the retrieval context length increases beyond 32k. Both models perform better at a 32k retrieval context length compared to using the entire 128k context window without RAG, with Qwen2.5 showing a notable improvement of +4.1%. In contrast, only GPT-4o effectively leverages longer retrieval context lengths, achieving the best RAG performance at 128k, while still lagging behind its overall score without RAG (-0.6%). These findings suggest that Qwen2.5 and GLM-4-Plus fall short in effectively utilizing and reasoning with information in context windows longer than 32k compared to GPT-4o. In addition, these experiments also confirm that the questions in LongBench v2 are challenging and cannot be solved solely through retrieval.

4.3 Measuring Memorization of Context

For an effective long-context benchmark, it is essential to ensure that LLMs cannot rely solely on memorizing previously seen data to answer questions. This necessitates the models to actively read and comprehend the provided long material in order to solve the problems. Following Bai et al. (2024b), we also evaluate the models' performance when providing only the questions, without the accompanying long context. The performance comparison between with (w/) and without (w/o) the context is presented in Table 3. As shown, without context,

most models achieve an overall accuracy ranging from 25% to 30%, which is comparable to random guessing. When comparing scores across different tasks, the memorization effect appears minimal for tasks II, III, and VI. The models perform best without context on tasks I and V, likely because they may have seen some of the documents, novels, or code repositories during training.

5 Conclusion

Our work introduces LongBench v2, a challenging multitask benchmark for long-context understanding and reasoning, carefully annotated and reviewed by human experts. LongBench v2 presents an equal challenge to both humans and state-of-the-art AI systems, with human performance at 50.1% and the best LLM achieving 57.7% accuracy, providing a reliable evaluation standard for the development of future superhuman AI systems. Our evaluation results also bring forward insights into the impact of scaling inference-time compute and RAG in long-context reasoning.

Limitations

We acknowledge certain limitations in our work, which we outline below: 1. **Benchmark size**: The benchmark's size may not be sufficiently large. While this can be seen as an advantage for quick evaluation, it could also lead to less stable results that are more vulnerable to randomness. Due to resource constraints, we are unable to expand the dataset at this time. Collecting the current 503 high-quality samples cost us 100,000 CNY and took more than two months. 2. **Language**: The current dataset is limited to English only. As a result, our benchmark does not yet capture the performance of models across multiple languages. 3. **Length distribution inconsistencies**: The length distribution across different tasks is uneven, with certain tasks concentrated around specific lengths. These differences in task distributions across length ranges make it difficult to provide a fair comparison of a single model's performance across length intervals. We recommend conducting comparisons between models on a per-interval basis. For instance, model A may outperform Model B in the short length range, while model B may outperform model A in the long length range. This would suggest that model B is better at handling longer tasks than model A.

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Appendix

A Author Contributions

- Project lead: YB
- Benchmark design: YB, ST, JZ, HP, XW, XL, SC
- Annotation platform: ST, YB
- Annotator recruitment: YB, JX, ST, JZ
- Annotator management: YB, ST, JZ, HP, XL, SC
- Evaluation: YB, assisted by JZ, XL
- Writing: YB, ST, assisted by JZ, HP, XW, XL
- Supervision and fundraising: JL, LH, JT, YD

B Task Descriptions

I.1. Single-Document QA (Academic)

Task Description: Ask questions based on academic articles (papers, textbooks), excluding content related to charts and figures within the text.
Example Questions: 1. Which methods were used to collect data in the study? 2. In what ways does the author's argument align or conflict with the findings of Smith et al. (2020)?

I.2. Single-Document QA (Literary)

Task Description: Ask questions about literary works, potentially covering characters, plot, writing style, and central themes.
Example Questions: 1. What are the key traits that define [character]'s personality? 2. What is the turning point in the novel, and how does it impact the characters? 3. What message does the author seem to be conveying through the ending?

I.3. Single-Document QA (Legal)

Task Description: Ask questions based on legal documents, referencing scenarios like legal consultations, case analysis, or legal document review.
Example Questions: 1. What is the basis of the defendant's defense? 2. How is the estate distributed according to the will? 3. What are the conditions for tax incentives mentioned in this regulation?

I.4. Single-Document QA (Financial)

Task Description: Ask questions based on financial documents, including but not limited to financial report analysis, market analysis, investment strategies, and risk assessment.
Example Questions: 1. Based on the report, how do changes in operational expenses align with the company's revenue growth strategy? 2. What macroeconomic indicators are likely to impact the company's performance in the next fiscal year, and how are they addressed in the document? 3. How does the document evaluate the impact of regulatory changes on the company's capital structure?

I.5. Single-Document QA (Governmental)

Task Description: Ask questions based on government reports and official documents, potentially covering policies, regulations, and public facilities.
Example Questions: 1. What are the main allocations for healthcare in this year's government budget? 2. Who qualifies for the education grants mentioned in this document? 3. How does this policy address the concerns of small businesses?

I.6. Single-Document QA (Detective)

Task Description: Ask questions based on a detective or mystery novel. Questions must be inferable after reading most of the novel, such as who the murderer is or what the method of the crime was, without the full reasoning or answer being directly present in the text.
Example Questions: 1. Who murdered Mary?

I.7. Single-Document QA (Event ordering)

Task Description: Given a long text (usually a novel) and 4 plot events from the novel in random order, the model is required to select the correct sequence of the plot development.
Example Questions: 1. Order the four events in their original order...

II.1. Multi-Document QA (Academic)

Task Description: Ask questions based on academic articles (papers, textbooks), excluding content related to charts and figures. Questions must require using the information from at least 2 documents to be answered, with no irrelevant documents.
Example Questions: 1. What are the improvements of the method in paper A compared with paper B?

II.2. Multi-Document QA (Legal)

Task Description: Ask questions based on legal documents, requiring at least 2 documents. Questions must require information from each document to be answered, and there should be no irrelevant documents.
Example Questions: 1. Is Zhang's crime a case of imagined concurrence or statutory concurrence of crimes?

II.3. Multi-Document QA (Financial)

Task Description: Ask questions based on financial documents, requiring at least 2 documents. Questions must require information from each document to be answered, and there should be no irrelevant documents.
Example Questions: 1. How has the R&D investment of the enterprises changed in the past ten years?

II.4. Multi-Document QA (Governmental)

Task Description: Ask questions based on government reports and official documents, requiring at least 2 documents. Questions must require information from each document to be answered, and there should be no irrelevant documents.
Example Questions: 1. How do the public transportation policies outlined in the 2022 Urban Development Report align with the environmental sustainability goals stated in the 2023 National Green Initiative document?

II.5. Multi-Document QA (Multi-news)

Task Description: Ask questions based on news articles, requiring at least 2 articles. Questions must require synthesizing information from multiple documents to be answered, and there should be no irrelevant documents.
Example Questions: 1. How have the top three positions in the medal leaderboard for the 2024 Paris Olympics changed over time?

III.1. Long In-context Learning (User guide QA)

Task Description: Given a long user guide, e.g., electronic device manual, software manual, musical instrument tutorial, annotate questions that require a deep understanding of the long text.
Example Questions: 1. I want to do time-lapse photography, how do I shoot it? 2. In what situations is it more effective to use parfor in MATLAB? 3. How can you change the timbre and achieve different expressive styles by controlling the force and speed of your key presses?

III.2. Long In-context Learning (New language translation)

Task Description: Translation tasks involving the rare languages Zhuang (vocabulary book and translation corpus from Zhang et al. (2024a)) and Kalamang (vocabulary book and translation corpus from Tanzer et al. (2024)), requiring reading a vocabulary book to complete.
Example Questions: 1. Translate the following kalamang into English: Wa me kariak kaia kon untuk emumur kalo tumun amkeiret mu wara nanet.

III.3. Long In-context Learning (Many-shot learning)

Task Description: Given many-shot examples, answer the query based on the given examples. All label information is anonymized and can only be learned from the examples. This task primarily involves multi-class classification datasets, including the named entity recognition dataset FewNERD (Ding et al., 2021), the relation classification dataset DocRED (Yao et al., 2019), the event detection dataset MAVEN (Wang et al., 2020), and the sentiment classification dataset GoEmotions (Demszky et al., 2020).
Example Questions: 1. What is the entity type of "glucagon"? 2. What is the relation type between "The Bone Forest" and "Robert Holdstock"? 3. What is the event type of "became"? 4. What are the emotions of the document "I'm more interested in why there are goldfish in the picture..."?

IV.1. Long-dialogue History Understanding (Agent history QA)

Task Description: Based on the agent dialogue history as context, ask questions about the content of the history. Specifically, we provide annotators with LLMs' dialogue history on playing games, which is derived from the GAMA-Bench (tse Huang et al., 2025). This dataset includes eight classical multi-agent games categorized into three groups: Cooperative Games, Betraying Games, and Sequential Games. In our task, we use them as context and annotate questions for the agent interaction history.
Example Questions: 1. Which player is the most selfish one in the fourth round of the game?

IV.2. Long-dialogue History Understanding (Dialogue history QA)

Task Description: Given a multi-turn chat history between a user and an AI assistant, raise a question than demands understanding the dialogue history. To ensure the length of the history, we sample data from LongMemEval (Wu et al., 2025a), which consists of over 500 sessions for each chat history that challenges the long-term memory capabilities of LLMs. We take the chat history as context and raise new questions for long-dialogue understanding.
Example Questions: 1. How long have I been living in my current apartment in Shinjuku?

V.1. Code Repository Understanding (Code repo QA)

Task Description: Based on a specific branch or commit of a codebase, annotate questions that require careful reading of multiple parts of the code or a deep understanding of the code's content to answer.
Example Questions: 1. For the current Megatron-LM framework, if I want to use the THD data format while enabling Context Parallel, how should I modify the experiments for rotary_pos_embedding?

VI.1. Long Structured Data Understanding (Table QA)

Task Description: Given a long table (e.g., financial report) or several interconnected tables, annotate questions that require integrating multiple cells or combining information from multiple tables. We provide annotators with long tables from the dataset proposed by TableLLM (Zhang et al., 2024c).
Example Questions: 1. For the industry fields involving entertainment, which grows most largely from 2021 to 2023?

VI.2. Long Structured Data Understanding (Knowledge graph reasoning)

Task Description: Given a large-scale knowledge graph, annotate questions and corresponding answers that require integrating multiple entities. We construct the knowledge graph (extracted from Wikidata (Vrandečić and Krötzsch, 2014)) and the complex logical queries based on the KQAPro dataset (Cao et al., 2022). Groundtruth answers are automatically derived by running the corresponding KoPL program (Cao et al., 2022; Yao et al., 2023) on the graph.

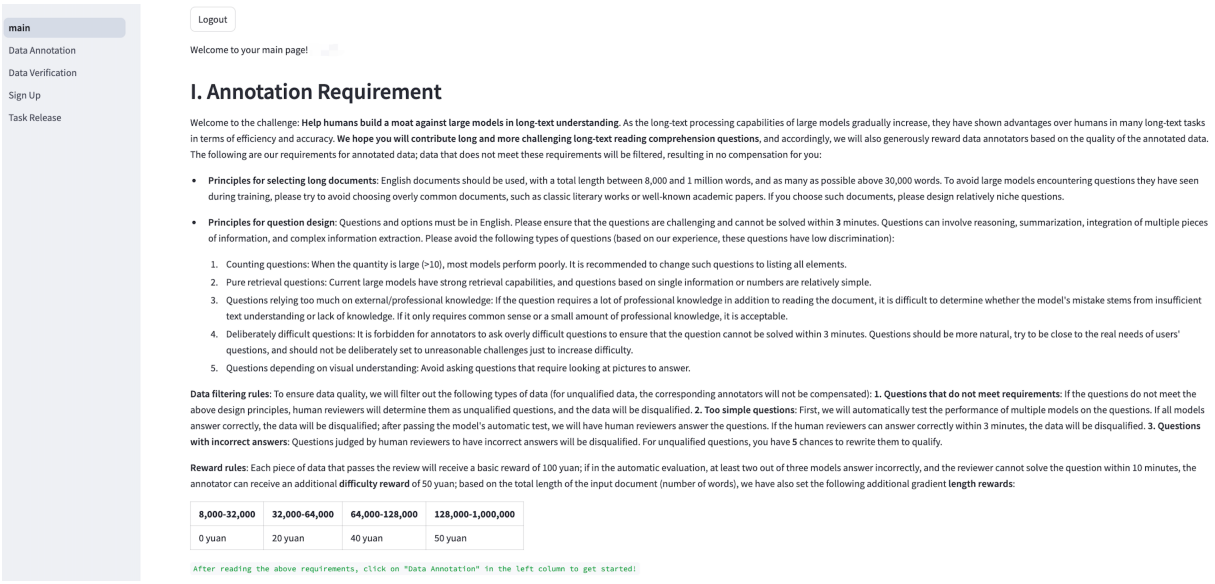


Figure 5: Screenshot of the main page (top part). After logging in, the annotator will first see this page, which displays our requirements and incentive policies. Annotators can also see the statuses of their data on this page.

Example Questions: 1. When did the people who captured Q10549 come to the region where Q231 is located?

C Annotation Details

C.1 Annotation Platform

Our annotation platform includes three pages: main page, data annotation page, and data verification page.

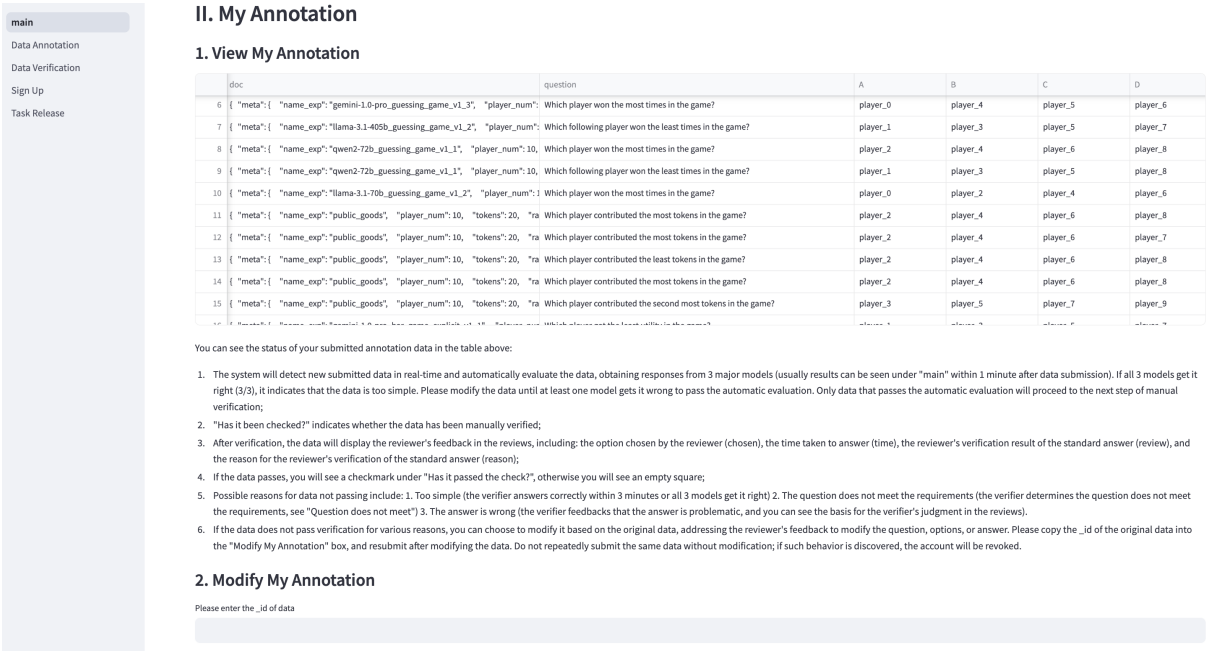


Figure 6: Screenshot of the main page (bottom part). Annotators can view the status of their data on this page. They can modify their rejected data for resubmission.

Main page. The main page serves as the central hub of the website, providing an overview of the tasks and data. Figure 5 shows the top part of the main page, where we display the annotation requirements for our task, allowing users to understand the demand of our annotation task. The bottom part of the main

page, as shown in Figure 6, also includes functionality to view the data status, where the feedback from automated and manual reviews is displayed. It also handles the deletion and modification of data. Each user can only view their own data and is not able to access others.

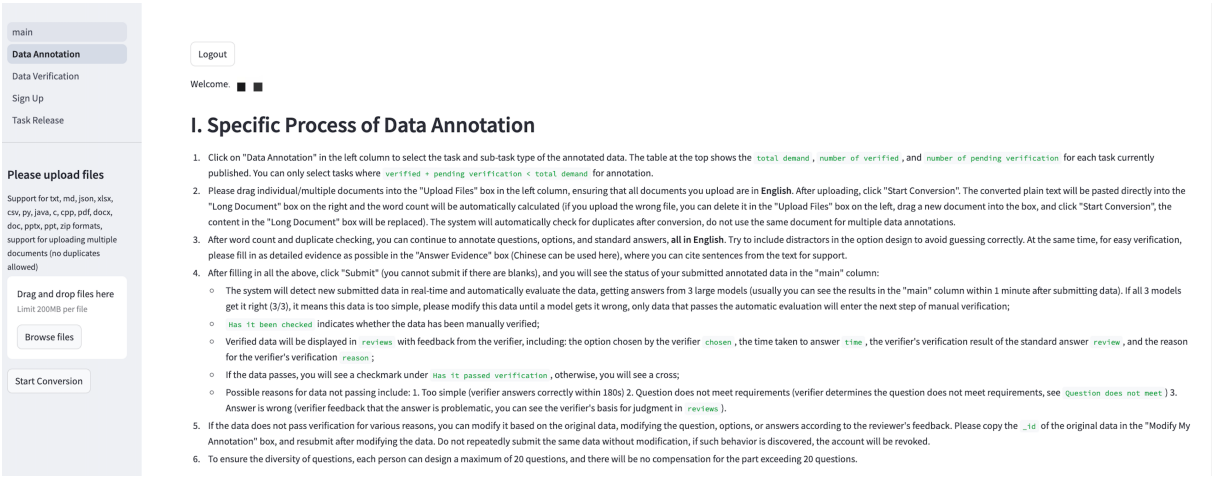


Figure 7: Screenshot of the data annotation page (top part).

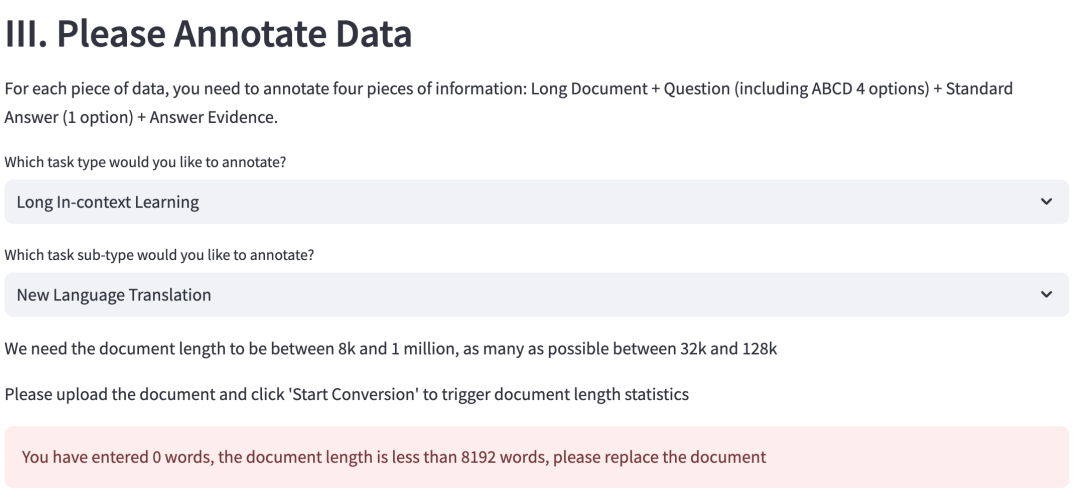


Figure 8: Screenshot of the data annotation page (bottom part). Annotator first uploads the document(s) and proposes a multiple-choice question based on the content.

Data annotation page. This page is designed for users to annotate long-context QA data. As shown in Figure 7, our guideline instructs users through the process of selecting tasks and subtasks, uploading documents, and annotating questions, options, and answers. The page ensures that all annotations are in English and meet specific requirements to challenge LLMs. As shown in Figure 8, annotators will first choose the task category they would like to annotate, then upload their documents to annotate a multiple-choice question. Our platform includes features to check for the word count and duplicate documents to ensure the length and diversity of documents. After questions are annotated, we conduct automated reviews to verify the complexity of the questions to ensure they are not overly simple. The page also provides instructions for annotating data and limits the number of questions each user can annotate to maintain diversity.

Data verification page. As illustrated in Figure 9, the data verification page is where human experts review the annotated data for accuracy and quality. Reviewers can only verify data that has passed the automated review and cannot verify data annotated by themselves. The page requires reviewers to download the documents and submit their own choice, and provide feedback on the correctness of the groundtruth answers. As shown in Figure 10, this page also allows users to flag questions that do not meet

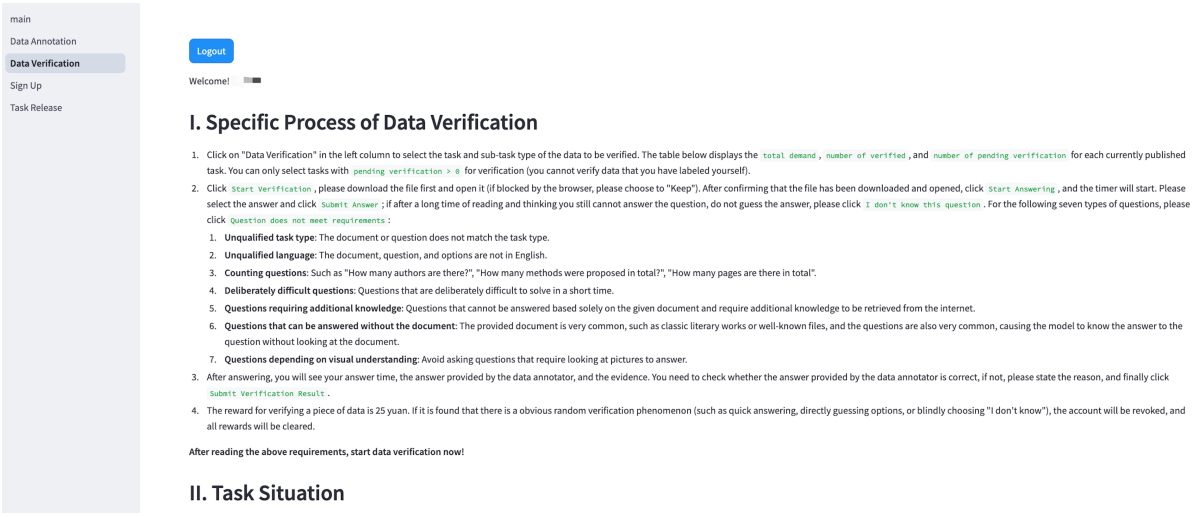


Figure 9: Screenshot of the data verification page (requirements part). Manual review will be conducted on this page to check whether the annotated data aligns with our requirements.

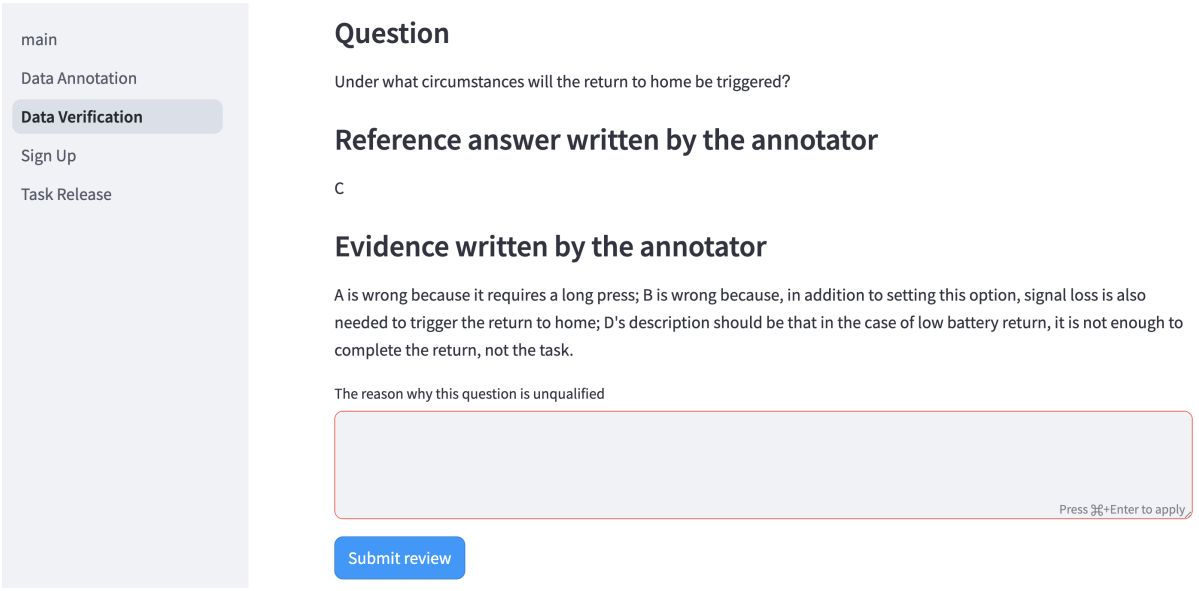


Figure 10: Screenshot of the data verification page after clicking the "Question does not meet requirements" button. Reviewers will use this page to write rejecting reasons if they decide that this question is unqualified.

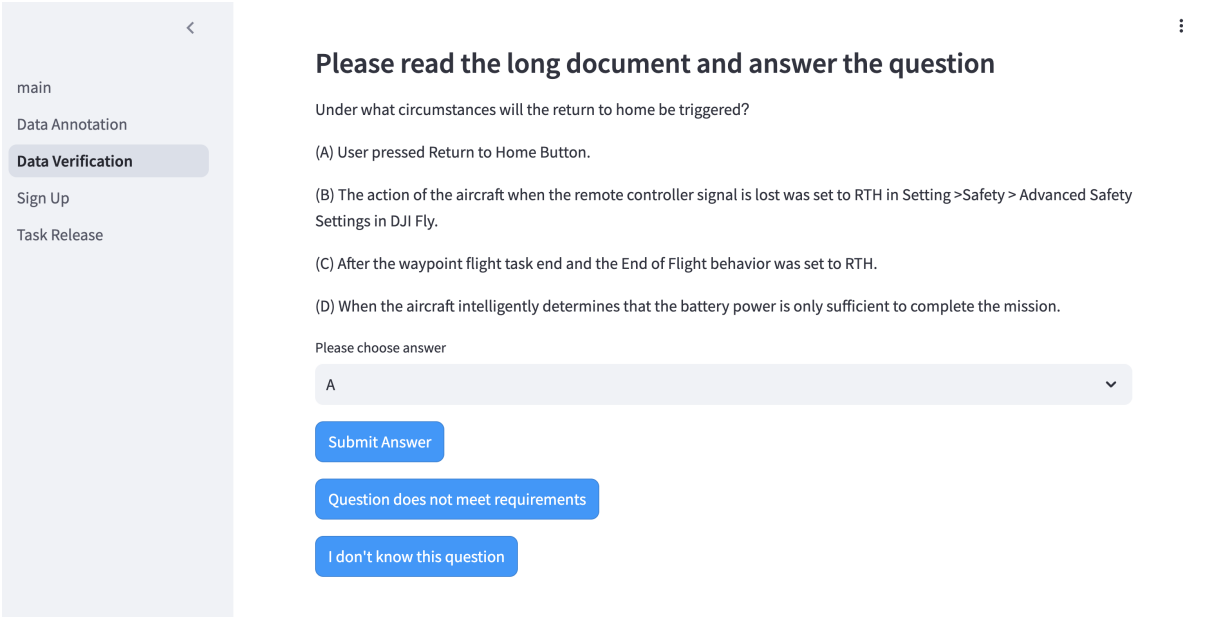


Figure 11: Screenshot of the data verification page for solving the question. Reviewers will enter this page when they attempt to answer the question. The long documents were downloaded before they answer the question.

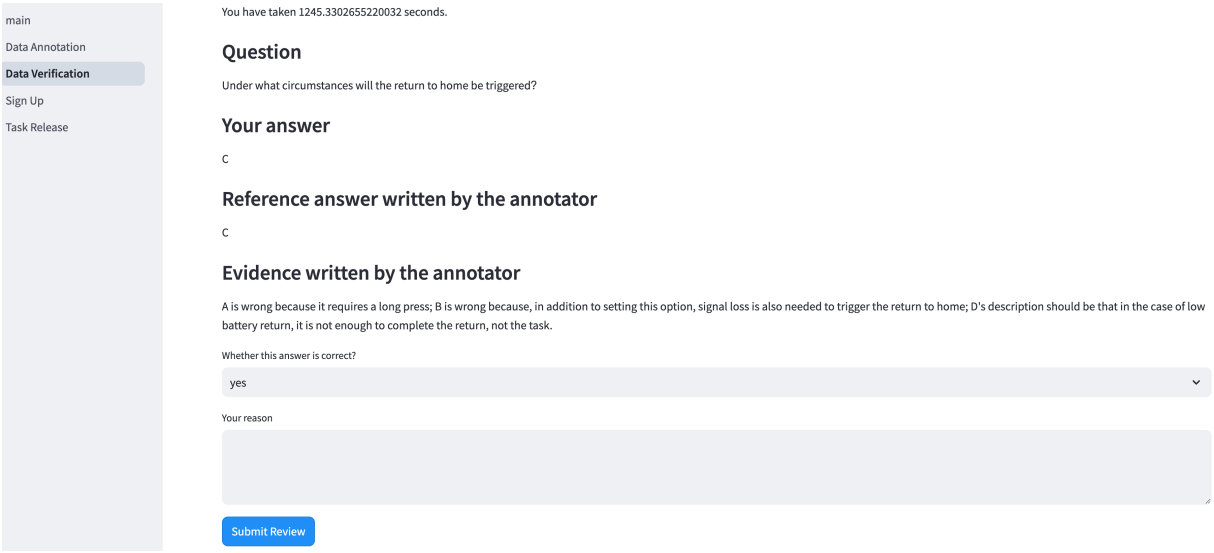
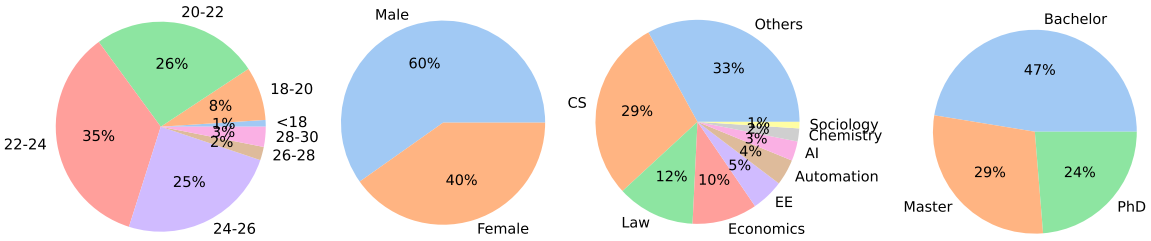


Figure 12: Screenshot of the data verification page after clicking the "Submit Answer" button. Reviewers will use this page to check whether the reference answer is correct and submit their reason.

the requirements, such as those that do not match the task type, or require additional knowledge beyond the provided document. If the question is qualified, then the reviewer will attempt to answer it, as shown in Figure 11. This process includes a timer to track the time taken to answer each question. Figure 12 shows the page when the reviewer finishes answering the question. The reviewer will be able to read the answer and evidence written by the annotator. The reviewer may check whether the answer is correct and submit the reason.

C.2 Annotator Statistics

To understand how diverse and professional our annotators are, we ask our annotators to fill in their age, gender, major, and degree during registration. We have ensured that no personal privacy information is leaked. Figure 13 displays the diverse distribution of annotators across various dimensions. In terms of age, the majority of annotators fall within the 20-22 (26%), 22-24 (35%), and 24-26 (25%) age groups because almost all annotators are recruited from universities. The distribution of majors is sufficiently diverse, with Computer Science (CS) being the most common (29%), followed by Law (24%) and Economics (22%). Finally, the majority of annotators are holding or pursuing a Bachelor's degree (47%), with a smaller proportion holding a Master's (29%) or PhD (24%). Each annotator can annotate at most 20 data to ensure the diversity of the data.



(a) Age (b) Gender (c) Major (d) Degree

Figure 13: Distribution of our annotators across ages, genders, majors, and degrees.

C.3 Annotation Guidelines

Overall annotation and platform guideline, displayed on the main page:

Welcome to the challenge: **Help humans build a moat against AI systems in long-context understanding**. As the long-context processing capabilities of large language models gradually increase, they have shown advantages over humans in many long-context tasks in terms of efficiency and accuracy. We invite you to contribute long and challenging long-context reading comprehension questions, and accordingly, we will also generously reward data annotators based on the quality of the annotated data. The following are our requirements for annotated data; data that does not meet these requirements will be filtered, resulting in no payment:

- **Principles for selecting long documents**: English documents should be used, with a total length between 8,000 and 2 million words, and as many as possible above 32,000 words. To avoid large language models encountering questions they have seen during training, please try to avoid choosing overly common documents, such as classic literary works or well-known academic papers. If you choose such documents, please design relatively niche questions.

- **Principles for question design**: Questions and options must be in English. Please make sure that the questions are challenging enough and cannot be solved within **3** minutes. Questions can involve reasoning, summarization, integration of multiple pieces of information, and complex information extraction. Please avoid the following types of questions (based on our experience, these questions have low discrimination):

1. *Counting-type questions*: When the quantity is large (>10), most models perform poorly. It is recommended to change such questions to listing all elements.
2. *Retrieval-type questions*: Current large language models have strong retrieval capabilities, and questions based on single information located somewhere in the document are relatively simple.
3. *Questions that rely too much on external/professional knowledge*: If the question requires a lot of professional knowledge in addition to reading the document, it is difficult to determine whether the model's mistake stems from insufficient text understanding or lack of knowledge. It is acceptable if it only requires common sense or a small amount of professional knowledge.
4. *Deliberately difficult questions*: It is forbidden for annotators to ask deliberately difficult and stilted questions just to ensure that the human reviewer cannot solve them within a short amount of time. Questions should be more natural, try to be close to the real needs of users' questions, and should not be deliberately set to unreasonable challenges just to increase difficulty.
5. *Questions that depend on visual understanding*: Avoid asking questions that require looking at pictures to answer.

Data filtering rules: To ensure data quality, we will filter out the following types of data (for unqualified data, the corresponding annotators will not be rewarded, and you have 5 chances to rewrite them to qualify):
1. *Questions that do not meet requirements*: If the questions do not meet the above requirements, human reviewers will determine them as unqualified questions, and the data will be disqualified.
2. *Too simple questions*: First, we will automatically test the performance of three models on the questions. If all models answer correctly, the data will be disqualified; after passing the model's automatic test, we will have human reviewers answer the questions. If the human reviewers can answer correctly within 3 minutes, the data will be disqualified.
3. *Questions with incorrect answers*: Questions judged by human reviewers to have incorrect answers will be disqualified.

Reward rules: Each piece of data that passes the review will receive a basic reward of `100 CNY`; if in the automatic evaluation, at least two out of three models answer incorrectly, and the reviewer cannot solve the question within 10 minutes, the annotator can receive an additional **difficulty reward** of `50 CNY`; based on the total length of the input document (number of words), we have also set the following additional stepped **length rewards**:
8,000 - 32,000 words: `0 CNY`
32,000 - 64,000 words: `20 CNY`
64,000 - 128,000 words: `40 CNY`
128,000 - 1,000,000 words: `50 CNY`

After reading the above requirements, click on "Data Annotation" in the left column to get started!

Guidelines provided to the annotators, displayed on the data annotation page:

1. Click on "Data Annotation" in the left column to select the task and subtask type of the annotated data. The table at the top shows the "total demand", "number of verified", and "number of pending verification" for each task. You can only select tasks where "verified + pending verification < total demand" for annotation.
2. Please drag individual/multiple files into the "Upload Files" box in the left column. Make sure that all files you upload are in **English**. After uploading, click "Start Conversion". The converted plain text will be pasted directly into the "Long Document" box on the right and the word count will be automatically calculated. If you upload the wrong file, you can delete it in the "Upload Files" box on the left, drag a new document into the box, and click "Start Conversion", the content in the "Long Document" box will be replaced. The system will automatically check for duplicates after conversion,

do not use the same document for multiple submissions.
3. After passing word counting and duplicate checking, you can continue to annotate questions, options, and answers, all in English. Try to include distractors in the option design to avoid guessing correctly. At the same time, for ease of verification, please fill in as detailed evidence as possible in the "Evidence" box, where you can cite sentences from the long context for support.
4. After filling in all the above, click "Submit" (you cannot submit if there are blanks), and you will see the status of your submitted annotated data in the "main" column:
- The system will detect newly submitted data in real-time and automatically evaluate the data, getting answers from 3 large language models (usually you can see the results in the "main" column within 1 minute after submitting data). If all 3 models get it right (3/3), it means this data is too simple, please modify this data until at least one model gets it wrong, only data that passes the automatic evaluation will enter the next step of manual verification.
- "`Checked?`" indicates whether the data has been manually verified.
- Verified data will be displayed in "`reviews`" with feedback from the verifier, including the option chosen by the verifier ("`chosen`"), the time taken to answer ("`time`"), the verifier's verification result of the groundtruth answer ("`correctness`"), and the reason for the verifier's judgment ("`reason`").
- If the data passes, you will see a checkmark under "`Verification passed?`", otherwise, you will see a cross.
- Possible reasons for data not passing include: (1). Too simple (3/3 models get it right or verifier answers correctly within 180s); (2). Question does not meet requirements (verifier determines the question does not meet requirements, see the "`reason`" box for the detailed reason); (3). The answer is wrong (you can see the verifier's basis for judgment in "`reason`").
5. If the data does not pass verification for various reasons, you can modify it based on the original data, modifying the question, options, or answer according to the reviewer's feedback. Please copy the "`_id`" of the original data in the "`Modify My Annotation`" box, and resubmit after modifying the data. Do not repeatedly submit the same data without modification, if such behavior is discovered, the account will be revoked.
6. To ensure the diversity of questions, each user can design a maximum of 20 questions.

Guidelines for the reviewers, displayed on the data verification page:

1. Click on "Data Verification" in the left column to select the task and subtask type of the data to be verified. The table below displays the "total demand", "number of verified", and "number of pending verification" for each current task. You can only select tasks with "pending verification > 0" for verification (you cannot verify data that you have labeled yourself).
2. Click "Start Verification", please download the file first and open it (if blocked by the browser, please choose "Keep"). After confirming that the file has been downloaded and opened, click "Start Answering", and the timer will start. Please select the answer and click "Submit Answer"; if after a long time (>15 min) of reading and thinking you still cannot answer the question, do not guess the answer, please click "I don't know the answer". For the following seven types of questions, please click "Question does not meet requirements": (1) *Mismatched task type*: The document or question does not match the task type. (2) *Unqualified language*: The document, question, and options are not in English. (3) *Counting questions*: Such as "How many authors are there?", "How many methods were proposed in total?", "How many pages are there in total". (4) *Deliberately difficult questions*: Questions that are deliberately difficult to solve in a short time. (5) *Questions requiring additional knowledge*: Questions that cannot be answered based solely on the given document and require additional knowledge to be searched from the internet. (6) *Questions that can be answered without the document*: The provided document is very common, such as classic literary works or well-known files, and the questions are also very common, causing the model to know the answer to the question without looking at the document. (7) *Questions depending on visual understanding*: Questions that require looking at visual contents to answer.

3. After answering, you will see your answer time, the answer provided by the data annotator, and the evidence. You need to check whether the answer provided by the data annotator is correct, if not, please fill in the reason, and finally click "Submit Verification Result".
4. The reward for verifying a piece of data is 25 CNY. If it is found that there is a malicious verification pattern (such as quick answering, directly guessing options, or blindly choosing "I don't know the answer"), the account will be revoked, and all rewards will be cleared.
After reading the above requirements, start data verification now!

C.4 Data Collection Cost

We spend approximately 100,000 CNY on data collection.

D More Evaluation Details

D.1 Baseline Models

Our open-source baselines include: GLM-4-9B-Chat (GLM et al., 2024), Llama-3.1-8B-Instruct, Llama-3.1-70B-Instruct, Llama-3.3-70B-Instruct (Dubey et al., 2024), Llama-3.1-Nemotron-70B-Instruct (Wang et al., 2025), Qwen2.5-7B-Instruct, Qwen2.5-72B-Instruct (Yang et al., 2024), Mistral-Large-Instruct-2407, Mistral-Large-Instruct-2411 (Jiang et al., 2023), and c4ai-command-r-plus-08-2024 (Cohere For AI, 2024). Our proprietary baselines include: GLM-4-Plus (GLM et al., 2024), GPT-4o-mini-2024-07-18 (OpenAI, 2024a), GPT-4o-2024-08-06, GPT-4o-2024-11-20 (OpenAI, 2024c), o1-mini-2024-09-12 (OpenAI, 2024d), o1-preview-2024-09-12 (OpenAI, 2024b), Claude-3.5-Sonnet-20241022 (Anthropic, 2024), Gemini-Exp-1206, and Gemini-2.0-Flash-Exp (Reid et al., 2024). All of the models mentioned above have a context window length of 128k tokens, with the exception of Claude-3.5-Sonnet-20241022 (200k), Gemini-Exp-1206 (2M), and Gemini-2.0-Flash-Exp (1M).

D.2 Evaluation Setting

In the zero-shot evaluation setting, we set the generation sampling parameters to `temperature`=0.1 and `max_new_tokens`=128. In the zero-shot + CoT setting, for the first model call where the model generates the chain-of-thought, we set `temperature`=0.1 and `max_new_tokens`=1024. For the subsequent model call where the model outputs the final answer, we set `temperature`=0.1 and `max_new_tokens`=128.

D.3 Evaluation Prompts

Prompt for zero-shot setting.

Please read the following text and answer the question below.

<text>
{*Long Context*}
</text>

What is the correct answer to this question: {*Question*}
Choices:
(A) {*Choice A*}
(B) {*Choice B*}
(C) {*Choice C*}
(D) {*Choice D*}

Format your response as follows: "The correct answer is (insert answer here)".

Prompt for zero-shot + CoT setting.

Please read the following text and answer the question below.

<text>
{*Long Context*}
</text>

What is the correct answer to this question: {*Question*}
Choices:
(A) {*Choice A*}
(B) {*Choice B*}
(C) {*Choice C*}
(D) {*Choice D*}

Let's think step by step:

Please read the following text and answer the questions below.

The text is too long and omitted here.

What is the correct answer to this question: {*Question*}
Choices:
(A) {*Choice A*}
(B) {*Choice B*}
(C) {*Choice C*}
(D) {*Choice D*}

Let's think step by step: {*Chain of thought generated in the last response*}

Based on the above, what is the single, most likely answer choice? Format your response as follows: "The correct answer is (insert answer here)".

E Deferred Experimental Results

			Difficulty				Length (<32k; 32k-128k; >128k)					
Model	**Overall**		**Easy**		**Hard**		**Short**		**Medium**		**Long**	
`Qwen2.5-7B-Instruct`	27.0	29.8	29.2	30.7	25.7	29.3	36.1	35.6	23.7	26.5	18.5	26.9
+*YaRN*	**30.0**	**35.6**	**30.7**	**38.0**	**29.6**	**34.1**	**40.6**	**43.9**	**24.2**	**32.6**	**24.1**	**27.8**
`Qwen2.5-72B-Instruct`	39.4	38.8	**43.8**	42.2	36.7	36.7	44.4	**50.0**	34.0	28.8	41.7	39.8
+*YaRN*	**42.1**	**43.5**	42.7	**47.9**	**41.8**	**40.8**	**45.6**	48.9	**38.1**	**40.9**	**44.4**	**39.8**

Table 4: Qwen2.5 results (%) using YaRN on LongBench v2. Higher scores in **bold**.

Qwen2.5 Results Using YaRN. Following the guidelines provided in the model card on `https://huggingface.co/Qwen/Qwen2.5-72B-Instruct`, we evaluate using YaRN with a scaling factor of 4.0. The results are presented in Table 4. YaRN significantly enhances both models' long-context processing ability on LongBench v2, especially on test cases >32k lengths (Medium & Long). Additionally, we observe that YaRN has a larger impact on model performance under the CoT setting, though the underlying reasons for this remain unclear.

Compensated Results. The compensated results that account for invalid outputs are shown in Table 5. We can see that the proportion of invalid outputs is relatively small, and it does not affect the conclusions drawn from our experimental results.

					Difficulty				Length (<32k; 32k-128k; >128k)					
Model	**Overall**		**Invalid**		**Easy**		**Hard**		**Short**		**Medium**		**Long**	
Open-source models														
GLM-4-9B-Chat	30.4	32.2	0.8	5.6	31.1	36.6	30.0	29.5	34.0	36.2	30.0	31.9	25.2	26.2
Llama-3.1-8B-Instruct	31.0	30.5	3.8	0.4	32.0	36.5	30.3	26.8	37.6	34.4	27.9	31.7	25.9	21.5
Llama-3.1-70B-Instruct	31.7	36.6	0.2	1.8	32.3	36.3	31.3	36.8	41.2	45.6	27.4	34.1	24.1	26.9
Llama-3.3-70B-Instruct	31.0	36.6	4.6	1.8	35.8	38.5	28.0	35.5	39.9	45.6	27.0	33.4	24.1	28.2
Llama-3.1-Nemotron-70B-Instruct	31.8	37.2	3.2	8.2	33.6	39.5	30.7	35.9	40.4	47.8	28.0	32.1	25.0	29.9
Qwen2.5-7B-Instruct	28.9	30.0	7.4	0.8	31.5	31.0	27.3	29.4	39.0	35.7	25.5	26.7	18.8	27.1
Qwen2.5-72B-Instruct	**40.4**	**39.2**	4.0	1.6	**44.4**	43.0	**37.9**	36.8	**46.7**	**50.1**	**34.2**	29.4	**42.1**	**40.3**
Mistral-Large-Instruct-2407	30.9	34.5	16.9	3.6	34.9	35.4	28.4	33.9	37.8	41.7	25.6	31.6	29.9	28.2
Mistral-Large-Instruct-2411	35.7	41.0	5.4	5.6	40.1	**45.3**	33.0	**38.3**	43.3	47.9	31.7	**36.0**	31.0	39.1
c4ai-command-r-plus-08-2024	28.8	32.0	3.8	1.4	31.0	34.9	27.4	30.1	37.4	39.6	25.2	24.8	21.5	33.6
Proprietary models														
GLM-4-Plus	44.6	47.6	1.0	5.8	47.5	53.5	42.8	43.9	50.7	54.7	46.5	46.2	30.6	38.4
GPT-4o-mini-2024-07-18	29.8	32.6	2.0	0.8	31.8	32.8	28.5	32.5	32.5	35.1	29.0	31.7	26.6	30.1
GPT-4o-2024-08-06	50.2	51.3	0.2	0.4	57.4	58.2	45.7	47.1	53.5	53.9	52.4	50.8	40.2	47.9
gpt-4o-2024-11-20	47.4	51.7	5.6	1.2	52.9	54.7	44.0	49.8	50.1	60.1	48.5	48.7	40.7	43.8
o1-mini-2024-09-12	38.3	39.4	1.8	2.0	39.7	43.4	37.4	36.9	48.7	49.6	34.0	33.5	29.0	34.3
o1-preview-2024-09-12	**57.9**	**57.1**	0.8	3.4	**67.1**	**60.5**	**52.3**	**55.0**	**62.7**	**65.3**	**53.8**	**51.1**	**58.3**	**55.5**
Claude-3.5-Sonnet-20241022	44.4	50.4	13.9	14.9	51.7	59.6	40.0	44.8	49.2	56.0	41.9	46.5	41.7	49.1
Gemini-Exp-1206	49.3	53.1	0.0	2.2	52.9	61.9	47.1	47.7	53.9	56.1	47.1	50.1	45.8	53.7
Gemini-2.0-Flash-Exp	48.8	52.6	5.6	6.2	52.5	60.5	46.5	47.7	51.9	58.9	48.1	47.0	44.9	53.5
Human	55.7		8.2		100		28.4		49.3		60.3		57.2	

Table 5: Compensated results (%) on LongBench v2. Due to the model's occasional refusal to answer or errors in the answer format under our zero-shot prompting, which leads to the failure of parsing selected options, these cases are classified as *invalid* outputs (invalid output rate presented in the table). We account for such cases by applying a 25% accuracy rate, and the compensated results are shown in this table. We also apply this compensation method to human baselines for cases where the human response is "I don't know the answer".